



Machine Learning

Project: Predictions of the WNBA

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Domain Description

This project is a data mining case study in the context of professional basketball tournaments. Each season follows a two-stage structure: a **regular season**, where all teams compete to maximize the number of wins, and a **playoff phase**, where the top teams face each other in knockout series to determine the champion.

The dataset covers **10 seasons** and includes detailed statistics about **20 teams**, **893 players**, **57 coaches**, and **11 types** of individual awards won by players. These entities provide the foundation for the predictive tasks introduced later in the presentation.

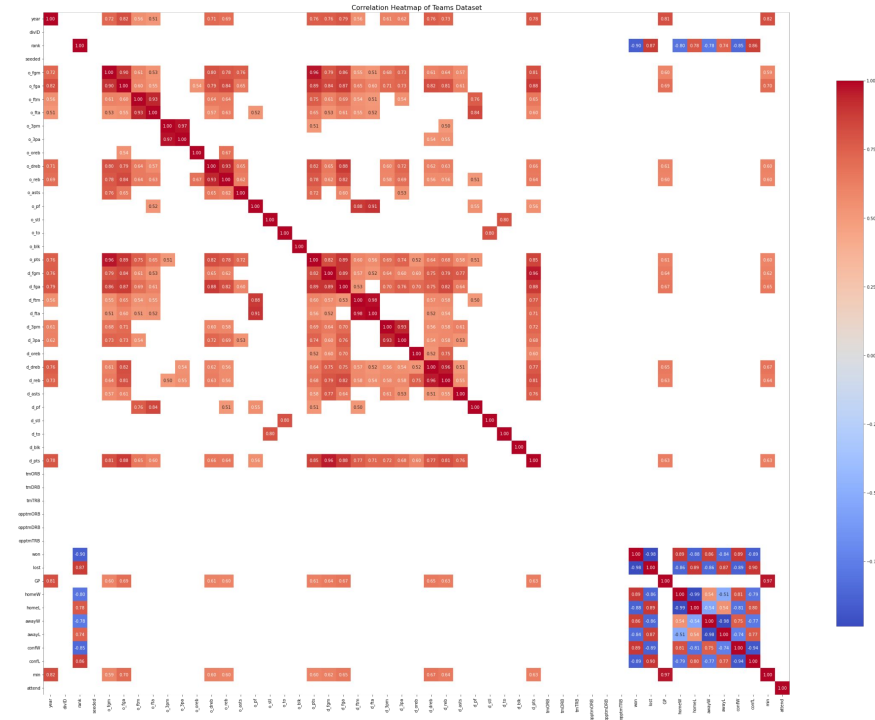
Beyond the core entity tables, three complementary relational datasets enrich the analysis:

- ***players_teams*** links each player to the team and season, allowing the evaluation of individual performance in a given year.
- ***teams_post*** summarizes team performance in the playoffs.
- ***series_post*** records the outcome of each playoff series, capturing the progression of teams through the postseason.

To understand the data structure and feature relationships, we performed correlation analysis on team statistics.

Key Findings:

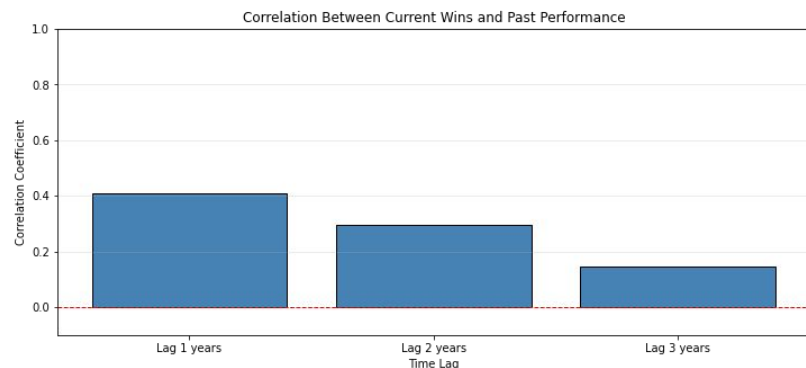
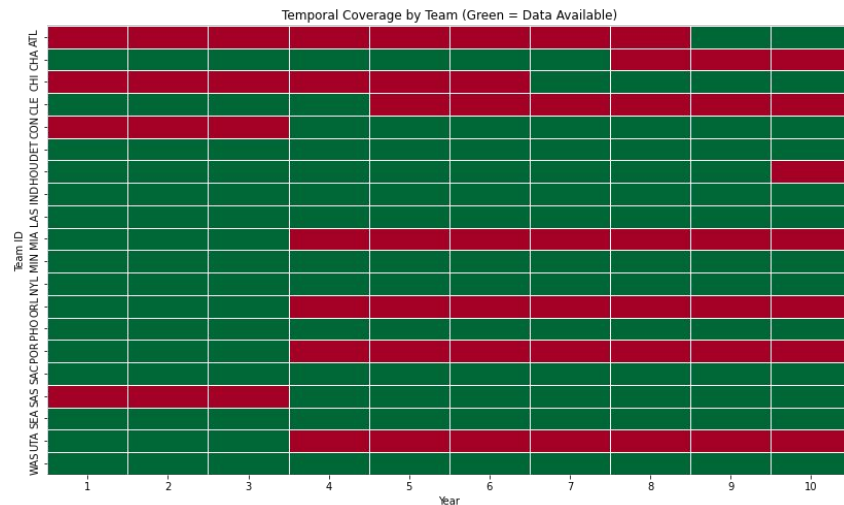
- **Strong positive correlation** among offensive metrics (field goals made, points, field goal attempts), confirming their interdependence.
- **Negative correlation** between team rank and offensive statistics, indicating that better-ranked teams naturally achieve superior offensive performance.
- **Rebound efficiency:** Offensive and defensive rebounds show positive correlation, suggesting that strong teams tend to excel in both dimensions.
- **Temporal trend:** Offensive statistics exhibit strong positive correlation with year, indicating improving team performance across the 10-season period.



Exploratory Data Analysis #2

The **first chart** shows that not all teams participate every season. It also highlights that some franchises share the same organization ID (FranchID) but should be treated as **different teams**, such as **ORL** (years 1–3) and **CON** (years 4–10).

The **correlation chart** (the second one) indicates that performance metrics from the immediately preceding season are the most strongly related to current wins. This supports the idea that older historical statistics quickly lose predictive power, reinforcing the importance of using recent data.





Exploratory Data Analysis #3

Outliers were identified in all datasets using **boxplots** but were not removed, as they capture distinctive performances (e.g., top teams versus average teams).

Each dataset was checked for potential **data quality issues**: missing values (including missing years), duplicates, consistency constraints (e.g., wins + losses = games played), and validity of numerical ranges (no negative values where they are not meaningful).

Descriptive analysis was performed with histograms and summary statistics to understand the distribution of key features.

Temporal analysis was used to study how metrics such as playoff wins or points per game evolved across seasons at the team/player level.

Comparative analysis aggregated metrics by team to compare cumulative performance over the 10-year period.

Statistical tests and correlations, including **t-tests** to compare average performance between two groups (for example, playoff vs non-playoff teams) and **Chi-square** tests to assess associations between categorical variables (for example, franchise and playoff qualification), were applied to compare groups, test associations, and identify which relationships between performance metrics are statistically significant.

3D Visualizations: 3D plots were used to visualize three variables simultaneously, improving intuition about complex relationships and offering an additional perspective on multi-dimensional patterns alongside traditional 2D charts.



Problem #1: Predicting Rank -> Problem Definition

The goal of this problem is to predict each team's final regular-season rank within its conference for a given year, using only information from **previous seasons**, as if the prediction were made the day before the new season starts.

The problem is framed as a **supervised learning task** where historical team, player, coach and performance statistics are transformed into features to predict the end-of-season conference rank. This required **data preparation** and **feature engineering** to build meaningful inputs, followed by **model development** and **evaluation** with appropriate metrics.

From a machine learning perspective, this is a **regression problem** with an **ordinal target**: the model predicts a continuous value for each team's final rank, which is then converted into discrete conference positions (1st, 2nd, 3rd, etc.).

Accurate rank predictions support strategic planning, highlight over- and under-performing teams before the season starts, and provide useful insights for coaches and analysts.



Data Preparation #1

Data cleaning focused on the datasets used for ranking (teams, awards_players, players_teams, and coaches), including removing non-informative columns such as **lgID** and team fields with only zeros or NaNs (**divID**, **seeded**, **tmORB**, **tmDRB**, **tmTRB**, **opptmORB**, **opptmDRB**, **opptmTRB**), and filtering out players with zero minutes and coaches without any games in a given season so that only active and informative records were kept.

After that, a rich set of basketball-specific features was constructed:

- **Overall and contextual performance:** win percentage (overall, home, away), net rating, point and assist/steal/block differentials
- **Shooting and scoring efficiency:** field goal, three-point, free-throw and true shooting percentages, effective field goal percentage, points per possession, offensive and defensive efficiencies
- **Rebounding and possession control:** total, offensive and defensive rebounding percentages and differentials, turnover-related ratios (assist-to-turnover, steal/block-to-turnover), pace and possession estimates



Data Preparation #2

- **Award-based quality signals:** counts of player awards and “Coach of the Year” awards aggregated at team–season level as proxies for talent and coaching quality
- **Advanced metrics used in real-world analytics:** Four Factors (offensive, defensive, and net) and Pythagorean win percentage with win differential, capturing expected vs actual performance

After generating many candidate features, only a subset was selected based on **domain knowledge** and **simple statistical criteria**. Features that were direct transformations or near-duplicates of each other were avoided, and preference was given to those with a meaningful correlation with the target rank (above **0.3** in absolute value).

VIF was explored to check multicollinearity between the selected features, although its impact was limited given the **small shortlist**, and **RFE** was tested as a wrapper method to confirm that the retained features were among the most informative for rank prediction.



Experimental Setup

1. Dataset and features - Used the engineered team-level dataset (*features_for_ranking.csv*)
2. Temporal features - Created **lag-1** features and **3-year weighted rolling averages** per team
3. Time-based splits - Initially used a **fixed chronological split** with **train** (years 2-8), **validation** (year 9) and **test** (year 10). Then adopted an **expanding rolling-window** scheme across seasons (2->3, 2-3->4, ..., 2-8->9), always training on past seasons and validating on the following one
4. Baseline - Simple baseline using **last season's rank** as the prediction
5. Model and feature-set comparison - Tested several models across three feature sets on the **validation year (9)**
6. Hyperparameter tuning - Using the best feature set found, we used **GridSearchCV** with pipelines (scaling + model) and **3-fold CV** to find the best configuration based on **MAE**
7. Final test evaluation - Evaluated all tuned models on **year 10** and selected the final model using Test **MAE**, **RMSE** and **Spearman**
8. Ranking and business metrics - Converted predictions to **conference rankings** and assessed **NDCG**, playoff **Precision@4** and a **playoff confusion matrix**
9. Feature Importance - Understanding what drives rank predictions

NDCG Score: 0.9421 (Closer to 1.0 is better)

--- Business Metrics Analysis (Playoff Precision) ---
Conference EA: Playoff Precision = 50% (2/4)
Conference WE: Playoff Precision = 75% (3/4)

AVG Playoff Precision: 62.5%

Results #1

=== Biggest ranking errors (Year 10) ===

confID	tmID	actual_rank	predicted_rank_model	rank_error
EA	ATL	2	7	5
EA	CON	6	3	-3
EA	NYL	7	4	-3
EA	WAS	4	6	2
WE	PHO	1	3	2
WE	SAC	6	4	-2
EA	DET	3	2	-1
WE	SEA	2	1	-1
WE	LAS	3	2	-1
WE	SAS	4	5	1

Rank error summary (all teams):

count	13.000000
mean	0.000000
std	2.236068
min	-3.000000
25%	-1.000000
50%	0.000000
75%	1.000000
max	5.000000

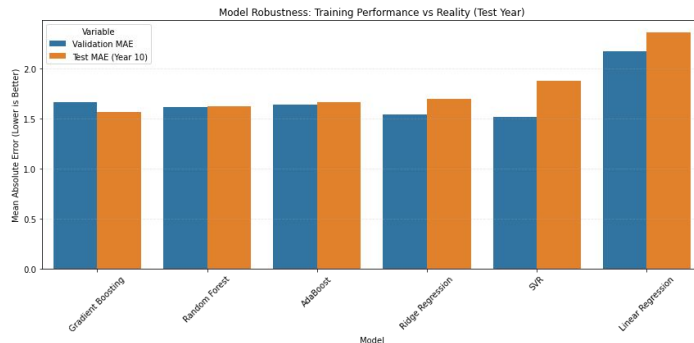
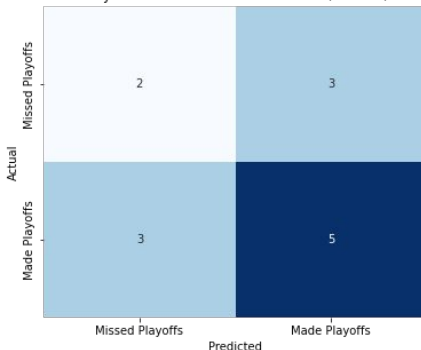
Rank error by conference:

confID	count	mean	std	min	25%	50%	75%	max
EA	7.0	0.0	2.828427	-3.0	-2.0	0.0	1.0	5.0
WE	6.0	0.0	1.549193	-2.0	-1.0	0.0	1.0	2.0

--- Final Showdown: Validation vs Test Performance ---

Model	Validation MAE	Test MAE (Year 10)	Test RMSE (Year 10)	Test Spearman	Generalization Gap
Gradient Boosting	1.662585	1.565960	1.949131	0.340732	-0.096625
Random Forest	1.616001	1.621726	2.013023	0.204993	0.005726
AdaBoost	1.636795	1.667556	2.014118	0.193913	0.030761
Ridge Regression	1.542538	1.692867	2.178466	0.152360	0.150329
SVR	1.512323	1.875235	2.450020	0.088646	0.362911
Linear Regression	2.176105	2.362403	3.035991	0.049863	0.186298

Playoff Prediction Confusion Matrix (Year 10)



--- Final Predicted vs. Actual Rankings (Year 10) ---

confID	tmID	actual_rank	baseline_rank	predicted_rank_model
EA	IND	1	4.0	1
EA	ATL	2	7.0	7
EA	DET	3	1.0	2
EA	WAS	4	6.0	6
EA	CHI	5	5.0	5
EA	CON	6	2.0	3
EA	NYL	7	3.0	4
WE	PHO	1	6.0	3
WE	SEA	2	2.0	1
WE	LAS	3	3.0	2
WE	SAS	4	1.0	5
WE	MIN	5	7.0	6
WE	SAC	6	4.0	4

--- Performance Comparison ---

Baseline Error (Previous Year Rank): 2.4615

Model Error (With Custom Features): 1.5660

TOTAL IMPROVEMENT: 36.38%

=== Weighted ensemble of top regression models (Year 10) ===

Ensemble MAE: 1.5984
Ensemble RMSE: 1.9948
Ensemble Spearman: 0.3158

--- Ensemble Predicted vs. Actual Rankings (Year 10) ---

confID	tmID	actual_rank	baseline_rank	predicted_rank_model
EA	IND	1	4.0	2
EA	ATL	2	7.0	7
EA	DET	3	1.0	1
EA	WAS	4	6.0	6
EA	CHI	5	5.0	5
EA	CON	6	2.0	3
EA	NYL	7	3.0	4
WE	PHO	1	6.0	3
WE	SEA	2	2.0	1
WE	LAS	3	3.0	2
WE	SAS	4	1.0	5
WE	MIN	5	7.0	6
WE	SAC	6	4.0	4

Results #2

```
--- Rolling-Window Models: Validation vs Test Performance ---
      Model  Validation MAE (rolling)  Test MAE (Year 10)  Test RMSE (Year 10)  Test Spearman
AdaBoost      0.976533              1.544050              2.000845              0.144248
Random Forest (no scaler)  1.437806              1.622920              2.041722              0.127428
Random Forest (scaled)    1.436613              1.624270              2.043744              0.127428
SVR              1.259536              1.649335              2.050289              0.166211
Ridge Regression  1.561943              1.677407              2.146349              0.185602
Gradient Boosting  0.973535              1.911375              2.358352              -0.069254
Linear Regression  4.324980              2.362403              3.035991              0.049863

Best rolling-window model on Test (Year 10): AdaBoost

--- Final Predicted vs. Actual Rankings (Year 10) - Rolling Model ---
confID tmID  actual_rank  baseline_rank  predicted_rank_model
EA  IND         1         4.0              3
EA  ATL         2         7.0              7
EA  DET         3         1.0              1
EA  WAS         4         6.0              6
EA  CHI         5         5.0              5
EA  COW         6         2.0              2
EA  NYL         7         3.0              4
WE  PHO         1         6.0              3
WE  SEA         2         2.0              2
WE  LAS         3         3.0              1
WE  SAS         4         1.0              5
WE  MIN         5         7.0              6
WE  SAC         6         4.0              4
```

Based on the previous slide, different experimental scenarios were explored, first comparing multiple models and then selecting the best one to evaluate with ranking and business metrics. Using the **fixed chronological split**, the best single model on the test season (Year 10) was **Gradient Boosting**, with a Test MAE of approximately **1.56**, clearly improving over the baseline that simply reused last season's rank.

On top of this, additional ranking-quality and business-oriented metrics were applied. An **NDCG** of **0.9421** indicates that the model is particularly accurate at identifying the top teams in each conference. The **average Playoff Precision@4** is **62.5%**, obtained by combining both conferences, with better performance in the Western Conference (**75%**) than in the Eastern Conference (**50%**). This pattern is consistent with the **error analysis**: the standard deviation of rank error is lower in the West (**1.54**) than in the East (**2.82**), suggesting more irregular and harder-to-predict outcomes in the Eastern Conference.

To go beyond a single-model view, a **weighted ensemble** of the top regression models was also tested. The ensemble achieved a slightly higher MAE of around **1.59**, which is expected given that it aggregates models that individually perform worse than the single best model.

Finally, in addition to the fixed train-validation-test split, an **expanding rolling-window** scheme across seasons was applied. Under this setup, the best model on the test year became **AdaBoost**, with a slightly lower Test MAE of about **1.54** compared to the best model obtained with the fixed-split approach, showing that the rolling-window strategy can yield a more robust configuration when respecting temporal structure more rigorously.



Problem #2: Predicting Coach Change

Problem definition:

The goal of this problem is to predict whether a coach will leave a given team or not next season. This is formulated as a **binary classification problem**, where the target is whether a coach departs (yes/no) based on historical team and coach performance metrics. Coach changes can occur for multiple reasons: poor results, personal reasons, or opportunities elsewhere. The prediction leverages patterns in teams' and coaches' previous performance to estimate the probability of a coach change.

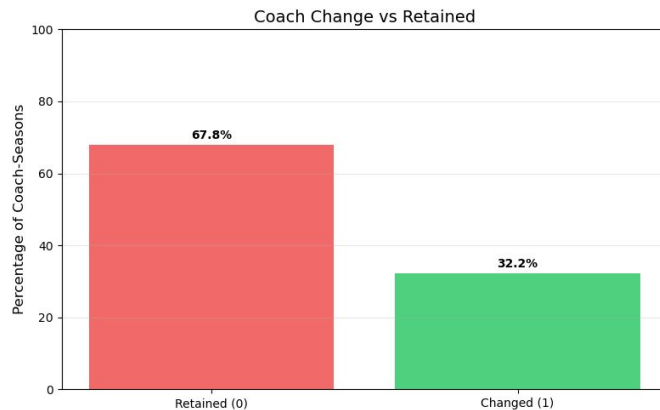
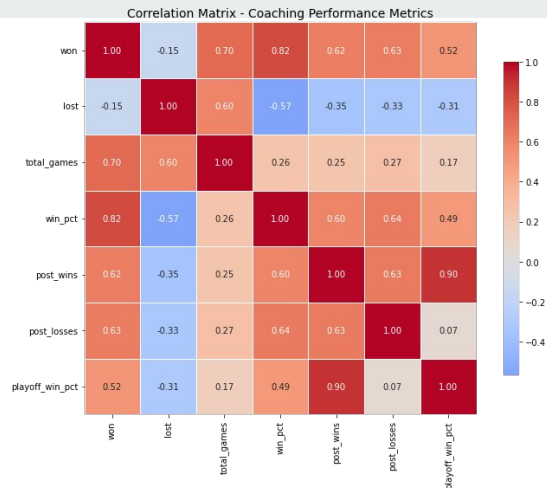
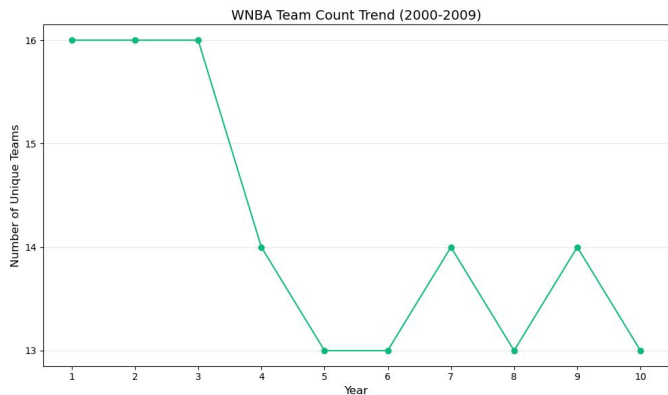
Key domain considerations:

- Not every team participates in every year
- There might be interim coaches
- Coach changes are often caused by disappointing results
- However, it could also happen due to other reasons: e.g. personal or financial reasons
- Multiple teams had to fold or relocate their teams during the test years
- Some coaches in the dataset had 0 games
- League status, WNBA was a new league at the time of 2000

Exploratory Data Analysis

Essential numbers found:

- 13-16 teams every year
- Total rows in coaches.csv: 163
- Unique Coach-Season Sample: 143
- Total Coach Change Sample: 46
- 8 coaches left after ending rank 1-2, ex:
 - laimbbi01w - due to personal reasons
 - coopemi01w - NBA job
 - westhpa01w - NBA job





Data Preparation

Defining the Target & Handling Time-Series

- **Problem**: Originally 163 rows in coaches.csv but should be only be 143.
- **Cause**: Team might change coach mid-season or have two interim coaches.
- **Solution**: Filter out the coach with lowest stint so only one coach per year per team.

Feature Engineering

- 36 features within: (Performance, Playoff success, Team quality, Four Factor, Tenure, Trends, Context)

Filtering process:

1. Domain knowledge
2. Correlation (Threshold at 0.2)
3. Multicollinearity (VIF)
4. Domain knowledge (keep career_win_pct and def_four_factors despite high VIF)

Target value: coach_changed (binary: 0 or 1).

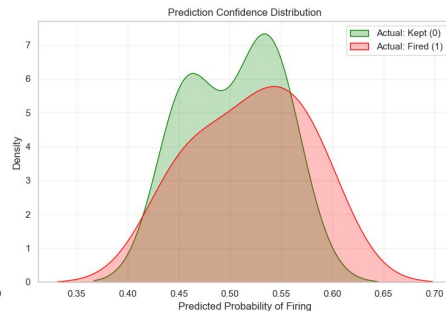
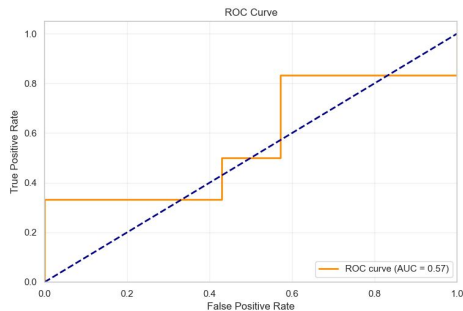


Experimental Setup

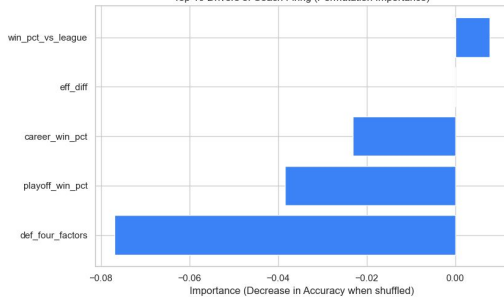
1. Data Cleaning
2. Data Preparation
3. Feature selection
4. Split training and test data
5. Training model
6. Prediction test with one model
7. Prediction with multiple models (choose best)
8. Cross-Validation and Single Holdout Validation
9. Ensemble Test

Results

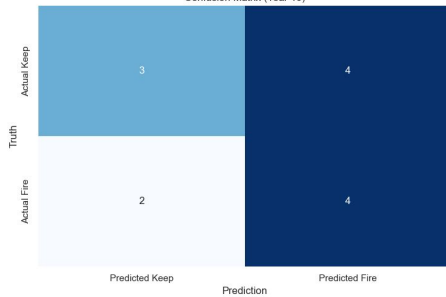
Coach Change Prediction Analysis (Year 10)
Model: LogisticRegression



Top 10 Drivers of Coach Firing (Permutation Importance)



Confusion Matrix (Year 10)

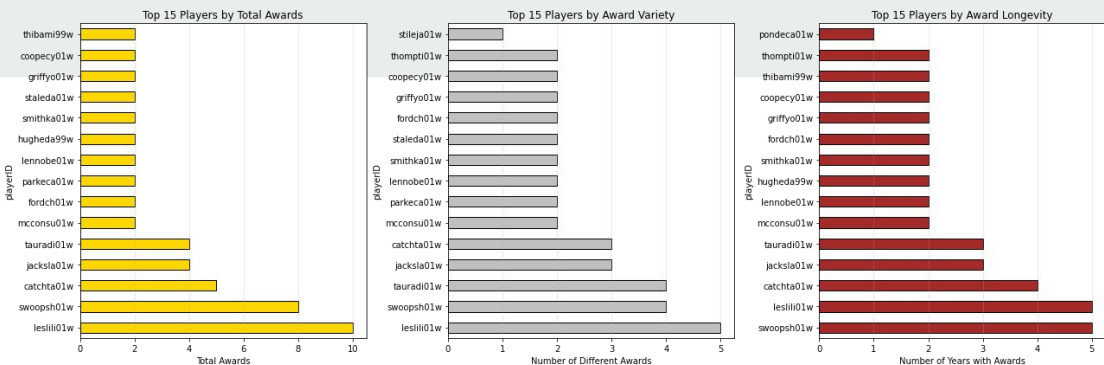


Rolling Validation Summary

Model	Avg Rolling F1	Avg Rolling AUC
Logistic Regression	0.594296	0.663559
Gradient Boosting	0.536686	0.654865
Random Forest	0.483364	0.639832
SVC	0.578423	0.619115
AdaBoost	0.436905	0.582962

Best Model: Logistic Regression

The best model (Logistic Regression) achieved an F1 score of **0.59** and AUC of **0.66**, outperforming random guessing despite inherent problem complexity. Coach departures are driven by factors beyond team performance metrics - including personal motivation, career opportunities in the NBA, and external circumstances. The exploratory analysis revealed that coaches with strong records still departed, and the **32.2%** coach change rate reflects this unpredictability. These metrics indicate that the model captures **some consistent patterns** in coaching transitions, even when the underlying phenomenon is partially stochastic and non-deterministic.



Problem #3: Predicting Awards

Problem definition:

Each award is treated as a separate prediction task for a given season: the model assigns a probability score to every eligible candidate. Higher probabilities indicate that the model considers the candidate more likely to win that award, and the candidate with the **highest** probability is the model's predicted winner (i.e., the favorite).

Exploratory Data Analysis

- 11 Different Awards
 - All-Star Game Most Valuable Player
 - Coach of the Year
 - Defensive Player of the Year
 - Kim Perrot Sportsmanship Award
 - Most Improved Player
 - Most Valuable Player
 - Rookie of the Year
 - Sixth Woman of the Year
 - WNBA Finals Most Valuable Player
 - WNBA All-Decade Team
 - WNBA All Decade Team Honorable Mention
- 54 Different Coaches
- 555 Different Players
- Each season, new players enter the league and others leave
- Player and coach movement between teams is frequent from year to year
- Not every award is presented every season, which creates gaps in the historical record
- Because awards are individual, injuries or sudden changes in role/minutes can strongly affect outcomes and reduce predictability
- Two awards (WNBA All-Decade Team and WNBA All-Decade Team Honorable Mention) were excluded due to insufficient data to train and evaluate reliable models

Experimental Setup



Even though each award was treated as an individual prediction task - requiring its own data preparation, exploratory analysis and feature engineering - the overall workflow and experimental structure remained essentially the same and is summarized below. Additional figures that support the analysis of specific awards are provided in the annexes.

1. Data Preparation

- Removed records that were irrelevant for prediction. The same baseline filtering was applied across all awards - for example, excluding players with no team assignment or with zero minutes played in a given season.

2. Feature Engineering

- Used **domain knowledge** and **data-driven analysis** to design and select features that match the specific characteristics of each award. For example, defensive-focused awards rely more on defensive indicators (e.g., rebounds and blocks), while other awards prioritize different performance and context variables.

3. Feature selection

- Evaluated the engineered features and retained the subset that provided the best predictive performance while minimizing redundancy. This selection combined **correlation-based filtering** (to remove highly correlated features) with **model-based validation** (e.g., importance and performance tests), tuned separately for each award.

4. Model Training

- Trained multiple classifiers - **Random Forest**, **XGBoost**, and **Logistic Regression** - and compared their performance per award. For Random Forest and XGBoost, we tested two variants (with and without feature scaling) to ensure preprocessing choices did not bias results, even though tree-based models are typically robust to scaling.
- Implemented **two time-based splitting strategies**: a fixed chronological split (train: Years 2-8, validation: Year 9, test: Year 10), and a rolling-window (walk-forward) validation approach to simulate repeated “predict next season” scenarios.
- For awards with very limited historical data (e.g., Sixth Woman with only a few seasons available), the validation window had to be shortened, which reduced the reliability of model selection and typically led to weaker results.

5. Evaluation

- After training, we evaluated each model by comparing its ranked predictions against the actual winner, where better performance means ranking the true winner as high as possible. We reported **Top-5 hit** and threshold-independent metrics (**ROC-AUC**, **Average Precision**) for consistent model comparison.

Most Improved Player Award

By the end of the data preparation, feature engineering and data analysis we ended up with the following features:

```
features_to_use = [  
    'prev_ppg', 'prev_mpg', 'prev_prod_per_minute',  
    'potential_ppg', 'breakout_score',  
    'age', 'experience', 'is_sophomore',  
    'vacated_min_pct', 'new_coach'  
]
```

The features with more importance for the train were the **'prev_ppg'**, **'prev_mpg'** and **'prev_prod_per_minute'** because they reflect the performance of the player in the last year. Although, we saw some standards in the players who won the award relatively to their age, experience and if is sophomore.

The **'vacated_min_pct'** are the minutes that will be “free” in a team when some important players change teams and usually, when a team change coaches its normal that the younger players have more opportunities to show themselves.

Overall ROC AUC: 0.6535

>>> Actual Winners & Model Prediction Ranks:

614	8	mccarja01w	5.0	0.267274	11.629454	14.033333
764	9	hoffmeb01w	11.0	0.046582	8.907216	17.117647
974	10	langhcr01w	4.0	0.075550	11.160681	15.558824

>>> Top 5 Predicted Candidates (Years 5-10):

	year	playerID	rank	prob	potential_ppg	vacated_min_pct
871	9	terryka01w	1.0	0.381347	12.043636	0.405732
776	9	joensca01w	2.0	0.345168	12.619926	0.261219
745	9	ferdima01w	3.0	0.286816	12.274428	0.555362
803	9	mazzake01w	4.0	0.282402	13.426230	0.295810
816	9	murphsh01w	5.0	0.132871	13.623853	0.292877
967	10	johnste01w	1.0	0.208460	8.793893	0.338412
1006	10	ohldeni01w	2.0	0.110786	12.686941	0.338412
943	10	gearlka01w	3.0	0.077417	13.153846	0.297103
974	10	langhcr01w	4.0	0.075550	11.160681	0.367033
1054	10	willili01w	5.0	0.075537	15.013100	0.207914

Results obtained using the **Random Forest** Algorithm. The Player who actually won was the **langhcr01w** and our model predict it in **4th place**.

Just a small number of features were strongly related to this problem and normally the winner was young and doesn't have to much stats to explore, what makes this prediction very difficult to make precisely.



Most Valuable Player Award

By the end of the data preparation, feature engineering and data analysis we ended up with the following features:

```
features = [  
    'prev_ppg', 'prev_rpg', 'prev_apg',  
    'prev_per_approx', 'prev_true_shooting',  
    'years_experience', 'height',  
    'team_rank_feature',  
    'score_rank',  
    'past_mvp_count',  
    'prev_ppg_rel', 'prev_per_approx_rel',  
    'prev_true_shooting_rel',  
    'usage_rate'  
]
```

We select features that best reflect the player's performance in the previous season, being mindful to exclude those with high correlations to avoid redundancy. Additionally, we observe that MVPs are almost exclusively players from teams that finish in the top 3. To enhance prediction accuracy, we incorporate our team ranking model from the first problem, which has proven to increase the likelihood of identifying the actual winner.

>>> Actual Winners & Model Prediction Ranks:

	year	playerID	rank	prev_ppg	prev_rpg	team_rank_feature
63	5	leslili01w	1.0	18.434783	10.043478	1
246	6	swoopsh01w	8.0	14.806452	4.935484	3
336	7	leslili01w	1.0	15.205882	7.294118	1
461	8	jacksla01w	2.0	19.500000	7.666667	4
821	10	tauradi01w	6.0	24.117647	5.058824	3

>>> Top 5 Predicted Candidates (Years 5-10):

	year	playerID	rank	prev_ppg	prev_rpg	team_rank_feature
604	9	jacksla01w	1.0	23.838710	9.677419	2
620	9	leslili01w	2.0	20.000000	9.500000	3
612	9	jonesas01w	3.0	15.300000	6.100000	2
681	9	swoopsh01w	4.0	7.666667	5.666667	2
682	9	tauradi01w	5.0	19.156250	4.218750	6
759	10	jacksla01w	1.0	20.238095	7.000000	1
795	10	parkeca01w	2.0	18.484848	9.484848	2
775	10	leslili01w	3.0	15.060606	8.878788	2
800	10	pondeca01w	4.0	21.218750	3.718750	3
732	10	douglka01w	5.0	15.636364	4.090909	1

Results obtained using the **Random Forest** Algorithm. The Player who actually won was the **tauradi01w** and our model predict it in **6th** place.

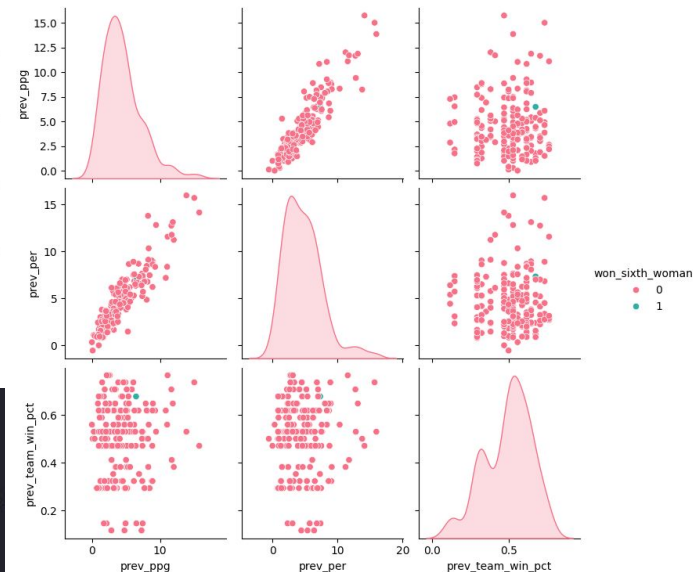
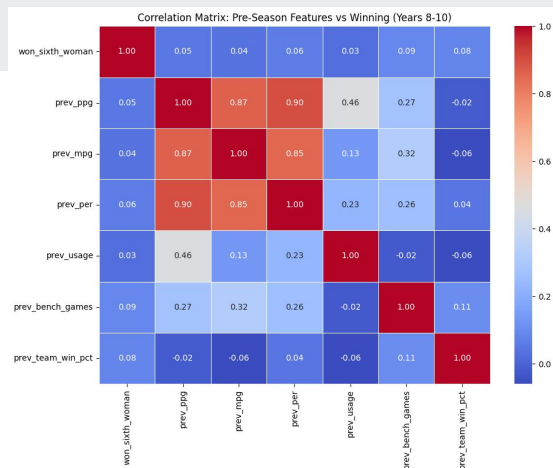
The large amount of relevant features available for this problem together with the integration of our ranking prediction enabled a really good prediction.

Sixth Woman Award

By the end of the data preparation, feature engineering and data analysis we ended up with the following features:

```
features_model = [
    'age', 'height',
    'coach_hist_bench_pct',
    'prev_ppg', 'prev_per', 'prev_bench_games',
    'prev_team_win_pct', 'prev_usage',
    'is_rookie'
] + college_features
```

Predicting this award proved to be extremely challenging due to the limited data, **only three years** in which the award was given. To complicate matters further, the winners in years 9 and 10 were **rookies**, for whom we had no available data. As a result, our model struggled to produce accurate predictions.



```
>>> Top Predictions Year 10:
      playerID tmID  prob  is_rookie  age  coach_hist_bench_pct  prev_ppg
261 hornba01w  DET  0.164         0    21         0.300000         5.411765
708 wiggica01w  MIN  0.144         0    19         0.300000        15.733333
730 wrighta01w  SEA  0.098         0    23         0.348762         7.882353
256 holliquo01w  MIN  0.078         1    18         0.300000         4.773460
36  battlas01w  NYL  0.058         0    24         0.300788         3.558824
581 smithbr01w  PHO  0.054         0    22         0.328157         4.100000
67  braxtko01w  DET  0.048         0    23         0.300000         8.939394
507 pierspl01w  DET  0.044         0    25         0.300000        11.857143
524 pricear01w  CHI  0.042         0    21         0.313164         6.911765
523 pricear01w  ATL  0.042         0    21         0.350251         6.911765
```

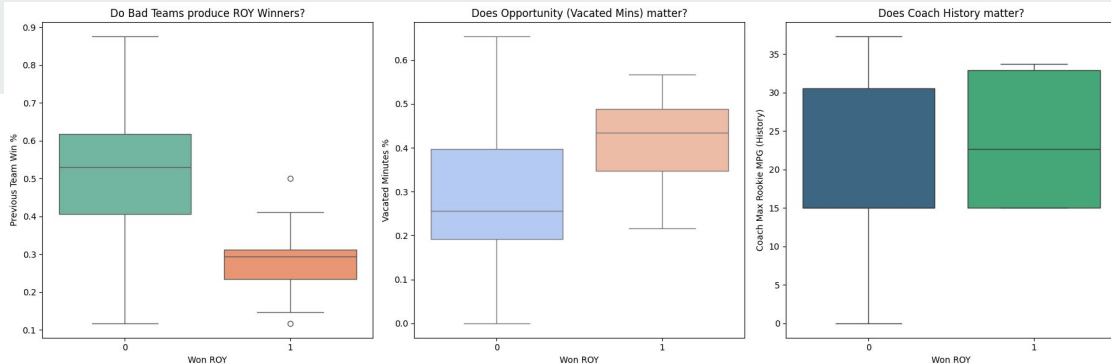
Rookie of the Year Award

By the end of the data preparation, feature engineering and data analysis we ended up with the following features:

```
features_enhanced = [  
    'prev_team_win_pct', 'vacated_min_pct',  
    'team_rank_feature', 'age', 'height',  
    'weight', 'pos_code'  
]  
features_enhanced.append('college_encoded')
```

Predicting this award was particularly challenging, as we had to forecast the winner without much prior information. To tackle this, we analyzed the relationship between team performance and the likelihood of a rookie player winning the award.

This approach revealed an important pattern: teams with **lower win percentages** tend to provide more opportunities for rookie players to take the spotlight and play significant roles, increasing their chances of winning. We also found a strong correlation with the **'vacated_min_pct'** metric, as it indicates more playing time for rookies, allowing them to showcase their talents and enhance their chances of winning the award.



Enhanced Prediction (Year 10)...

Top 5 Candidates (Year 10):

	playerID	tmID	prob	age	height	pos_simple	college_y
313	mccouan01w	ATL	0.148	20	73.0	F	Louisville
100	colemma01w	WAS	0.138	19	73.0	G	Maryland
481	tolivkr01w	CHI	0.128	19	67.0	G	Maryland
273	lehnish01w	ATL	0.078	20	69.0	G	Kansas State
332	montgre01w	MIN	0.028	20	67.0	G	Connecticut

Results obtained using the **Random Forest** algorithm showed strong accuracy. The player who ultimately won was **mccouan01w**, and our model correctly predicted them in **1st** place.

The key to achieving these accurate results lies in a thorough data analysis and careful selection of the most relevant features. Notably, features such as **college performance** and **team rank prediction** played a crucial role. By incorporating data from previous years, particularly team rankings and college-level performance, we were able to better assess a rookie's potential. Additionally, the training, validation, and testing methods used to fine-tune the model contributed significantly to its predictive power.

Kim Perrot

Sportsmanship Award

```
# Features that change every year (require history/lags)
dynamic_features = [
    'fouls_per_36', 'disqualifications', 'assist_to_turnover',
    'turnovers_per_game', 'turnovers_per_36', 'positive_actions',
    'in_team_ast_pct', 'in_team_to_pct', 'assists_per_game',
    'starter_ratio', 'stint', 'minutes_per_game', 'clean_defense'
]

# Features that are known/static at the start of the season
static_features = [
    'age_proxy', 'is_guard', 'team_tenure'
]
```

For the Kim Perrot Sportsmanship Award model, dynamic performance features were first standardized by season (**z-scores**) and then transformed into **Lag-1** values (last year's behavior) and **3-year weighted rolling averages** (recent trend), which were combined with z-scored static features like age proxy, position and team tenure before training. Z-scores were used instead of raw values because they express each player's stat relative to that season's league context - i.e., how many standard deviations above or below the season mean the player was - making comparisons across years more meaningful.

```
--- Grid Search on TRAIN only (years < 9) for 5 models ---

Training RandomForest_no_scaler...
Fitting 3 folds for each of 54 candidates, totalling 162 fits
> Best ROC AUC (CV on train): 0.8936
> Best Params: {'model__class_weight': 'balanced_subsample', 'model__max_depth': 10, 'model__min_samples_leaf': 4, 'model__n_estimators': 300}

Training RandomForest_scaled...
Fitting 3 folds for each of 54 candidates, totalling 162 fits
> Best ROC AUC (CV on train): 0.8936
> Best Params: {'model__class_weight': 'balanced_subsample', 'model__max_depth': 10, 'model__min_samples_leaf': 4, 'model__n_estimators': 300}

Training GradientBoosting_no_scaler...
Fitting 3 folds for each of 27 candidates, totalling 81 fits
> Best ROC AUC (CV on train): 0.9571
> Best Params: {'model__learning_rate': 0.05, 'model__max_depth': 2, 'model__n_estimators': 200}

Training GradientBoosting_scaled...
Fitting 3 folds for each of 27 candidates, totalling 81 fits
> Best ROC AUC (CV on train): 0.9571
> Best Params: {'model__learning_rate': 0.05, 'model__max_depth': 2, 'model__n_estimators': 200}

Training LogisticRegression_scaled...
Fitting 3 folds for each of 4 candidates, totalling 12 fits
> Best ROC AUC (CV on train): 0.9210
> Best Params: {'model__C': 10.0}
```

These ROC-AUC scores demonstrate that Gradient Boosting excels at ranking true award winners above non-winners (**AUC = 0.96**), while Logistic Regression performs very well (**AUC = 0.92**) and Random Forest also shows solid performance (**AUC = 0.89**). These results were obtained using the **fixed-split approach**. The **sliding-window** evaluation, presented in the next slide, yields consistently better AUC results and provides greater confidence in model reliability. Additionally, we explored **model ensembling** on the fixed split as a way to potentially improve performance, though it did not outperform individual models. Finally, **error analysis** was conducted to identify which player characteristics led the model to select a deteriorating player over the true winner, offering insights into model decision-making.

This award is inherently harder to predict from box-score stats because it recognizes **on-court sportsmanship** like ethical conduct, fair play and integrity, rather than pure performance. Even so, our feature analysis suggests a clear **experience signal**: winners tend to be older and more established (higher age proxy and longer team tenure relative to the league), which helps the model separate likely candidates.

Kim Perrot Sportsmanship Award

```
-- Validation performance (Year 9) for tuned models --
```

```
-- Model ranking on Validation (Year 9) --
```

	Model	Val AUC	Val AP	Top-5 hit (val)
	RandomForest_no_scaler	1.000000	1.0	1
	RandomForest_scaled	1.000000	1.0	1
	GradientBoosting_no_scaler	1.000000	1.0	1
	GradientBoosting_scaled	1.000000	1.0	1
	LogisticRegression_scaled	0.993333	0.5	1

Best model on validation: RandomForest_no_scaler

```
-- Validation Results (Year 9) --
```

Top 5 Favorites Predicted by Model:

playerID	tmID	is_kpsa	prob_winning
johnsvi01w	SAS	1	0.730069
beviltu01w	IND	0	0.255321
swoopsh01w	SEA	0	0.184333
mcwiltat01w	WAS	0	0.133690
mcwiltat01w	DET	0	0.081853

Actual Winner (johnsvi01w) Rank: #1

Winner Probability: 0.7301 (Top Favorite: 0.7301)

```
=== Soft-voting ensemble on Test (Year 10) - KPSA ===
```

Ensemble Test AUC: 0.8489

Ensemble Test AP: 0.0455

```
-- Ensemble Top-5 candidates (Year 10) --
```

playerID	tmID	is_kpsa	prob_winning
johnsvi01w	SAS	0	0.473876
beviltu01w	IND	0	0.454386
mcwiltat01w	DET	0	0.452372
harrokr01w	LAS	0	0.287435
griffyo01w	IND	0	0.284953

Ensemble Top-5 hit (test): 0

Ensemble: Actual winner lawsoka01w rank (Year 10): #22

Winner prob (ensemble): 0.0187 (Top prob: 0.4739)

```
-- Evaluating tuned models retrained on TRAIN+VAL --
```

```
-- Final comparison on Test (Year 10) --
```

	Model	Test AUC	Test AP	Top-5 hit (test)
	RandomForest_no_scaler	0.870504	0.052632	0
	RandomForest_scaled	0.870504	0.052632	0
	LogisticRegression_scaled	0.705036	0.023810	0
	GradientBoosting_no_scaler	0.683453	0.016129	0
	GradientBoosting_scaled	0.683453	0.016129	0

Best model on Test: RandomForest_no_scaler

```
-- TEST PREDICTION (Year 10) --
```

Model's Top 5 Candidates (RandomForest_no_scaler):

playerID	tmID	is_kpsa	prob_winning
johnsvi01w	SAS	0	0.265613
beviltu01w	IND	0	0.236276
mcwiltat01w	DET	0	0.230633
penicti01w	SAC	0	0.147455
griffyo01w	IND	0	0.091108

Actual Winner (lawsoka01w) Rank (Year 10): #19

Winner Probability: 0.0262 (Top Favorite: 0.2656)

```
-- Generalization gap (model chosen on validation) --
```

AUC gap (test - validation): -0.1295

AP gap (test - validation): -0.9474

```
-- Final comparison on Test (Year 10) using rolling-selected hyperparams --
```

	Model	wins	mean_auc_cv	Test AUC	Test AP
	RandomForest_no_scaler	2	0.932143	0.863309	0.050000
	RandomForest_scaled	2	0.932143	0.863309	0.050000
	GradientBoosting_no_scaler	1	1.000000	0.719424	0.017241
	GradientBoosting_scaled	1	1.000000	0.719424	0.017241
	LogisticRegression_scaled	4	0.869130	0.690647	0.022727

```
-- Head-to-Head: Actual Winner (lawsoka01w) vs Model Favorite (johnsvi01w) ---
Comparing feature inputs (Positive Diff = Favorite has higher value):
```

	WINNER (lawsoka01w)	FAVORITE (johnsvi01w)
>> PREDICTED PROBABILITY	0.0262	0.2656
z_fouls_per_36_lag1	-1.0682	-1.1053
z_disqualifications_lag1	-0.5027	-0.5027
z_assist_to_turnover_lag1	0.8021	2.3733
z_turnovers_per_game_lag1	0.2150	0.2534
z_turnovers_per_36_lag1	-0.5025	-0.5732
z_positive_actions_lag1	0.2906	1.4383
z_in_team_ast_pct_lag1	0.9868	1.6623
z_in_team_to_pct_lag1	0.3453	0.5658
z_assists_per_game_lag1	0.6299	1.8737
z_starter_ratio_lag1	1.4288	1.4288
z_stint_lag1	-0.3511	-0.3511
z_minutes_per_game_lag1	0.8614	0.9616
z_clean_defense_lag1	0.6874	0.7016
z_fouls_per_36_avg3yr	-0.9983	-1.0234
z_disqualifications_avg3yr	-0.4852	-0.4852
z_assist_to_turnover_avg3yr	0.7049	2.4441
z_turnovers_per_game_avg3yr	0.0739	0.1365
z_turnovers_per_36_avg3yr	-0.4506	-0.5164
z_positive_actions_avg3yr	0.2006	1.3355
z_in_team_ast_pct_avg3yr	0.8014	1.7136
z_in_team_to_pct_avg3yr	0.2676	0.4527
z_assists_per_game_avg3yr	0.4737	1.7821
z_starter_ratio_avg3yr	0.3031	1.3531
z_stint_avg3yr	-0.3001	-0.3001
z_minutes_per_game_avg3yr	0.7358	0.9661
z_clean_defense_avg3yr	0.6294	0.6625
z_age_proxy	0.3357	2.4042
z_is_guard	1.0658	1.0658
z_team_tenure	2.0769	0.6471

```
=== RandomForest no scaler: Top 10 predictions (Year 10) ===
```

playerID	tmID	is_kpsa	prob_winning
beviltu01w	IND	0	0.298957
mcwiltat01w	DET	0	0.291061
johnsvi01w	SAS	0	0.288171
griffyo01w	IND	0	0.112250
penicti01w	SAC	0	0.103338
smithka01w	DET	0	0.063677
perkijio01w	CHI	0	0.049068
leslili01w	LAS	0	0.049048
mitchle01w	NYL	0	0.048496
johnssh01w	SEA	0	0.044646

-> True winner lawsoka01w rank: #20 (prob=0.0222)

Defensive Player of the Year Award

```
selected_features = [
    "steals_per_game", "blocks_per_game", "def_reb_per_game",
    "def_stocks_per_game", "stocks_to_fouls_ratio", "minutes_per_game",
    "games_started_pct", "turnovers_per_game", "in_team_blk_pct",
    "in_team_stl_pct", "in_team_dreb_pct", "win_pct", "team_def_rank"
]
```

For the Defensive Player of the Year model, the same methodology was applied: defensive performance features were standardized by season (**z-scores**), converted into Lag-1 values (previous year's defensive profile) and 3-year weighted rolling averages (recent defensive trend), and then combined with season-level static attributes before training the classifier (**experience**).

```
--- Validation performance (Year 9) for tuned models ---

--- Model ranking on Validation (Year 9) ---
      Model Val AUC Val AP Top-5 hit (val)
XGBoost_no_scaler 0.993333 0.500000
XGBoost_scaled 0.993333 0.500000
LogisticRegression_scaled 0.993333 0.500000
RandomForest_no_scaler 0.986667 0.333333
RandomForest_scaled 0.986667 0.333333

Best model on validation: XGBoost_no_scaler

--- Validation Results (Year 9) ---
Model's Top 5 Predictions (best validation model):
  playerID tmID is_dpoj prob winning
catchta01w IND 0 0.809799
leslili01w LAS 1 0.053659
fordch01w DET 0 0.032781
beviltu01w IND 0 0.030045
jacksla01w SEA 0 0.028970

Year 9 Actual Winner (leslili01w) Rank: #2
```

```
=== Soft-voting ensemble on Test (Year 10) - DPOY ===
Ensemble Test AUC: 0.9784
Ensemble Test AP: 0.2500

--- Ensemble Top-5 candidates (Year 10) ---
  playerID tmID is_dpoj prob winning
jacksla01w SEA 0 0.290635
parkeca01w LAS 0 0.237395
leslili01w LAS 0 0.228255
catchta01w IND 1 0.154261
anosini01w MIN 0 0.143559

Ensemble Top-5 hit (test): 1

Ensemble: Actual DPOY catchta01w rank (Year 10): #4
Winner prob (ensemble): 0.1543 (Top prob: 0.2906)

--- Evaluating tuned models retrained on TRAIN+VAL ---

--- Final comparison on Test (Year 10) ---
      Model Test AUC Test AP Top-5 hit (test)
RandomForest_no_scaler 0.978417 0.25 1
RandomForest_scaled 0.978417 0.25 1
XGBoost_no_scaler 0.971223 0.20 1
XGBoost_scaled 0.971223 0.20 1
LogisticRegression_scaled 0.971223 0.20 1
```

```
--- Grid Search on TRAIN only (years < 9) for 5 models ---

Training RandomForest_no_scaler...
Fitting 3 folds for each of 54 candidates, totalling 162 fits
> Best ROC AUC (CV on train): 0.8275
> Best Params: {'model_class_weight': 'balanced_subsample', 'model_max_depth': 5, 'model_min_samples_leaf': 1, 'model_n_estimators': 100}

Training RandomForest_scaled...
Fitting 3 folds for each of 54 candidates, totalling 162 fits
> Best ROC AUC (CV on train): 0.8275
> Best Params: {'model_class_weight': 'balanced_subsample', 'model_max_depth': 5, 'model_min_samples_leaf': 1, 'model_n_estimators': 100}

Training XGBoost_no_scaler...
Fitting 3 folds for each of 32 candidates, totalling 96 fits
> Best ROC AUC (CV on train): 0.8379
> Best Params: {'model_colsample_bytree': 0.8, 'model_learning_rate': 0.03, 'model_max_depth': 3, 'model_n_estimators': 200, 'model_subsample': 0.8}

Training XGBoost_scaled...
Fitting 3 folds for each of 32 candidates, totalling 96 fits
> Best ROC AUC (CV on train): 0.8379
> Best Params: {'model_colsample_bytree': 0.8, 'model_learning_rate': 0.03, 'model_max_depth': 3, 'model_n_estimators': 200, 'model_subsample': 0.8}

Training LogisticRegression_scaled...
Fitting 3 folds for each of 4 candidates, totalling 12 fits
> Best ROC AUC (CV on train): 0.8913
> Best Params: {'model_C': 0.01}
```

Although Logistic Regression posted the best ROC-AUC in train cross-validation (**0.89**), Random Forest and XGBoost were the models that most consistently placed the true Defensive Player of the Year among the top-ranked candidates. Regarding the rolling-window evaluation, performance improved further - reaching an AUC as high as **0.987** in the final split - which makes the results more reliable because it better reflects year-to-year generalization. Overall, this award was more “learnable” than the previous ones, since DPOY is driven more directly by **observable defensive patterns** that a model can capture from statistics.

Best model on Test: RandomForest_no_scaler

```
--- TEST Results (Year 10) ---
Model's Top 5 Predictions (RandomForest_no_scaler):
  playerID tmID is_dpoj prob winning
jacksla01w SEA 0 0.394876
parkeca01w LAS 0 0.314731
leslili01w LAS 0 0.305088
catchta01w IND 1 0.196311
youngso01w SAS 0 0.180472
```

Year 10 Actual Winner (catchta01w) Rank: #4
Probability: 0.1963

```
--- Generalization gap (model chosen on validation) ---
AUC gap (test - validation): -0.0221
AP gap (test - validation): -0.3000
```

```
--- Final comparison on Test (Year 10) using rolling-selected hyperparams (DPOY) ---
      Model wins mean auc cv Test AUC Test AP
XGBoost_no_scaler 1 1.000000 0.992806 0.500000
XGBoost_scaled 1 1.000000 0.992806 0.500000
LogisticRegression_scaled 5 0.865892 0.971223 0.200000
RandomForest_no_scaler 1 1.000000 0.964029 0.166667
RandomForest_scaled 1 1.000000 0.964029 0.166667

=== XGBoost_no_scaler: Top 10 predictions (Year 10) ===
  playerID tmID is_dpoj prob winning
anosini01w MIN 0 0.086835
catchta01w IND 1 0.082388
parkeca01w LAS 0 0.073029
jacksla01w SEA 0 0.066272
leslili01w LAS 0 0.064091
fowlesy01w CHI 0 0.018943
youngso01w SAS 0 0.018004
hornba01w DET 0 0.015397
beviltu01w IND 0 0.011657
dougla01w IND 0 0.011394

-> True winner catchta01w rank: #2 (prob=0.0824)
```

WNBA Finals MVP Award

```
team_flag_features = [
    "is_champion",
    "team_made_playoffs",
    "team_made_finals",
    "team_playoff_win_pct"
]
```

For the Finals MVP award model, the same historical-features methodology was applied as for the other awards, but the input variables focused on **playoff per-game** statistics, so that the model emphasizes players who sustain high impact over heavy minutes for their team rather than normalizing everything to per-36 rates that can inflate the numbers of lower-minute bench players. Because Finals MVP candidates overwhelmingly come from teams that make a **deep playoff run** (and most often the eventual champion), we also introduced compact **team-context playoff signals** (e.g., made playoffs / reached the finals).

For the test season (Year 10) and the forecast season (Year 11), we used our **best team-ranking model** to filter candidates to players on **predicted top-4 teams per conference**, accepting the risk of missing an unexpected playoff team, since the ranking model was stable (**MAE < 1.6**) and highly accurate at identifying top teams (**NDCG = 0.9421**), which improved Finals MVP prediction in those years.

```
selected_features = [
    "playoff_points_per_game",
    "playoff_impact_per_game",
    "playoff_minutes_per_game",
    "playoff_games_started_pct",
    "playoff_rebounds_per_game",
    "playoff_assists_per_game",
    "playoff_steals_per_game",
    "playoff_blocks_per_game",
    "playoff_fg_pct",
    "win_pct"
]

target = "is_finals_mvp"
```

With only 7 positive examples in the training set, all models faced significant challenges in distinguishing true Finals MVP winners. Random Forest achieved the best ROC-AUC (**0.72**), marginally ahead of XGBoost (**0.68**) and Logistic Regression (**0.52**), indicating difficulty in consistently ranking the rare positive candidates above negatives. This inferior performance reflects a structural constraint: without a predictive filter to identify teams that actually reached the playoffs in those years, the model trained on heavily noisy candidate pools (**players from non-playoff teams**), making it harder to learn a clear pattern.

```
--- Grid Search on TRAIN only (years < 9) for 5 models ---
Positives in train: 7

Training RandomForest_no_scaler...
Fitting 3 folds for each of 54 candidates, totalling 162 fits
> Best ROC AUC (CV on train): 0.7211
> Best Params: {'model__class_weight': 'balanced_subsample', 'model__max_depth': None, 'model__min_samples_leaf': 4, 'model__n_estimators': 200}

Training RandomForest_scaled...
Fitting 3 folds for each of 54 candidates, totalling 162 fits
> Best ROC AUC (CV on train): 0.7211
> Best Params: {'model__class_weight': 'balanced_subsample', 'model__max_depth': None, 'model__min_samples_leaf': 4, 'model__n_estimators': 200}

Training XGBoost_no_scaler...
Fitting 3 folds for each of 32 candidates, totalling 96 fits
> Best ROC AUC (CV on train): 0.6841
> Best Params: {'model__colsample_bytree': 0.8, 'model__learning_rate': 0.1, 'model__max_depth': 5, 'model__n_estimators': 400, 'model__subsample': 0.8}

Training XGBoost_scaled...
Fitting 3 folds for each of 32 candidates, totalling 96 fits
> Best ROC AUC (CV on train): 0.6841
> Best Params: {'model__colsample_bytree': 0.8, 'model__learning_rate': 0.1, 'model__max_depth': 5, 'model__n_estimators': 400, 'model__subsample': 0.8}

Training LogisticRegression_scaled...
Fitting 3 folds for each of 4 candidates, totalling 12 fits
> Best ROC AUC (CV on train): 0.5242
> Best Params: {'model__C': 0.01}
```

WNBA Finals MVP Award

--- Validation performance (Year 9) for tuned models ---

--- Model ranking on Validation (Year 9) ---

Model	Val AUC	Val AP	Top-5 hit (val)
RandomForest_no_scaler	0.13	0.006623	0
RandomForest_scaled	0.13	0.006623	0
LogisticRegression_scaled	0.10	0.007353	0
XGBoost_no_scaler	0.06	0.007042	0
XGBoost_scaled	0.06	0.007042	0

Best model on validation: RandomForest_no_scaler

--- Validation Results (Year 9) ---

Model's Top 5 Predictions (best validation model):

playerID	tmID	is_finals_mvp	prob_winning
wilkibr01w	SAS	0	0.427616
queenbr01w	CHI	0	0.427616
terryka01w	ATL	0	0.244265
mosbybe01w	WAS	0	0.186392
bondla01w	IND	0	0.186034

Year 9 Actual FMVP (smithka01w) Rank: #142

=== DETAILED FEATURE COMPARISON - YEAR 9 ===

Model Top 1: wilkibr01w (SAS)

Model Top 2: queenbr01w (CHI)

Actual FMVP: smithka01w (DET)

FEATURE	TOP1	TOP2	FMVP
z_playoff_points_per_game_lag1	-0.62	-0.62	1.43
z_playoff_impact_per_game_lag1	-0.66	-0.66	1.31
z_playoff_minutes_per_game_lag1	-0.72	-0.72	2.00
z_playoff_games_started_pct_lag1	-0.54	-0.54	1.86
z_playoff_rebounds_per_game_lag1	-0.63	-0.63	0.56
z_playoff_assists_per_game_lag1	-0.58	-0.58	2.08
z_playoff_steals_per_game_lag1	-0.56	-0.56	0.63
z_playoff_blocks_per_game_lag1	-0.38	-0.38	-0.03
z_playoff_fg_pct_lag1	-0.73	-0.73	0.67
z_win_pct_lag1	0.65	1.47	1.66
z_playoff_points_per_game_avg3yr	-0.62	-0.62	1.28
z_playoff_impact_per_game_avg3yr	-0.66	-0.66	1.08
z_playoff_minutes_per_game_avg3yr	-0.72	-0.72	1.54
z_playoff_games_started_pct_avg3yr	-0.54	-0.54	1.38
z_playoff_rebounds_per_game_avg3yr	-0.63	-0.63	0.29
z_playoff_assists_per_game_avg3yr	-0.58	-0.58	1.83
z_playoff_steals_per_game_avg3yr	-0.56	-0.56	0.35
z_playoff_blocks_per_game_avg3yr	-0.38	-0.38	-0.21
z_playoff_fg_pct_avg3yr	-0.73	-0.73	0.56
z_win_pct_avg3yr	0.65	1.47	1.02
is_champion_lag1	0.00	0.00	0.00
team_made_playoffs_lag1	1.00	1.00	1.00
team_made_finals_lag1	1.00	0.00	1.00
team_playoff_win_pct_lag1	0.67	0.60	0.55
experience	1.00	1.00	9.00

--- Final comparison on Test (Year 10) using rolling-selected hyperparams (FMVP) ---

Model	wins	mean_auc_cv	Test AUC	Test AP
XGBoost_scaled	2	0.053333	0.733333	0.040000
XGBoost_no_scaler	2	0.053333	0.733333	0.040000
LogisticRegression_scaled	6	0.455956	0.633333	0.029412
RandomForest_no_scaler	1	0.864286	0.361111	0.016949
RandomForest_scaled	1	0.864286	0.361111	0.016949

--- Evaluating tuned models retrained on TRAIN+VAL ---

--- Final comparison on Test (Year 10) ---

Model	Test AUC	Test AP	Top-5 hit (test)
LogisticRegression_scaled	0.633333	0.029412	0
XGBoost_scaled	0.522222	0.022727	0
XGBoost_no_scaler	0.488889	0.021277	0
RandomForest_no_scaler	0.444444	0.019608	0
RandomForest_scaled	0.444444	0.019608	0

Best model on Test: LogisticRegression_scaled

--- TEST Results (Year 10 Forecast) ---

Model's Top 5 Predictions (LogisticRegression_scaled):

playerID	tmID	is_finals_mvp	prob_winning
leslilil01w	LAS	0	0.871779
catchta01w	IND	0	0.848532
birdsu01w	SEA	0	0.800254
jonesas01w	CON	0	0.793704
parkeca01w	LAS	0	0.788999

Year 10 Actual FMVP (tauradi01w) Rank: #34

Predicted Probability: 0.4671

--- Generalization gap (model chosen on validation) ---

AUC gap (test - validation): +0.3144

AP gap (test - validation): +0.0130

=== XGBoost_no_scaler: Top 10 predictions (Year 10) ===

playerID	tmID	is_finals_mvp	prob_winning
catchta01w	IND	0	0.160272
leslilil01w	LAS	0	0.029166
parkeca01w	LAS	0	0.023204
mcwiltat01w	DET	0	0.022766
taylope01w	PHO	0	0.018596
birdsu01w	SEA	0	0.017331
smithka01w	DET	0	0.016204
suttota01w	IND	0	0.013562
penicti01w	SAC	0	0.010773
miltode01w	LAS	0	0.010599

-> True winner tauradi01w rank: #25 (prob=0.0029)

=== Soft-voting ensemble on Test (Year 10) - Finals MVP ===

Ensemble Test AUC: 0.6333

Ensemble Test AP: 0.0294

--- Ensemble Top-5 candidates (Year 10) ---

playerID	tmID	is_finals_mvp	prob_winning
catchta01w	IND	0	0.401117
leslilil01w	LAS	0	0.344634
parkeca01w	LAS	0	0.332274
birdsu01w	SEA	0	0.311465
jonesas01w	CON	0	0.305758

Ensemble Top-5 hit (test): 0

Ensemble: Actual FMVP tauradi01w rank (Year 10): #34

Winner prob (ensemble): 0.1803 (Top prob: 0.4011)

Coach of the Year Award

```
selected_features = [  
    'win_percentage', 'stint', 'playoff_win_pct', 'made_playoffs',  
    'rank', 'confW', 'confL', 'conference_win_pct',  
    'homeW', 'homeL', 'home_win_pct', 'away_win_pct', 'home_advantage',  
    'offensive_efficiency', 'defensive_efficiency'  
]  
  
narrative_features = ['experience']
```

For the Coach of the Year model, the feature set focused mainly on **team-level performance** metrics (regular-season and playoff win percentages, home/away and conference records, offensive and defensive efficiency) complemented by an **experience** feature, reflecting the assumption that award-winning coaches are **strongly associated** with consistently high-performing teams.

```
--- Final comparison on Test (Year 10) using rolling-selected hyperparams (COY) ---  


|                           | Model | wins     | mean auc | cv       | Test AUC | Test AP |
|---------------------------|-------|----------|----------|----------|----------|---------|
| RandomForest_no_scaler    | 1     | 1.000000 | 1.0      | 1.000000 |          |         |
| RandomForest_scaled       | 1     | 1.000000 | 1.0      | 1.000000 |          |         |
| XGBoost_no_scaler         | 2     | 0.211111 | 1.0      | 1.000000 |          |         |
| XGBoost_scaled            | 2     | 0.211111 | 1.0      | 1.000000 |          |         |
| LogisticRegression_scaled | 4     | 0.450000 | 0.5      | 0.142857 |          |         |

  
=== RandomForest_no_scaler: Top 10 predictions (Year 10) ===  


| coachID     | tmID | won_coy | prob_winning |
|-------------|------|---------|--------------|
| meadoma99w  | ATL  | 1       | 0.316150     |
| whisejo99w  | SAC  | 0       | 0.187052     |
| bouceje01w  | SAC  | 0       | 0.182432     |
| thibami99w  | CON  | 0       | 0.165813     |
| donovan99w  | NYL  | 0       | 0.164107     |
| copleya99w  | NYL  | 0       | 0.150211     |
| hugheda99w  | SAS  | 0       | 0.133031     |
| coppenm01w  | LAS  | 0       | 0.125485     |
| gainesco01w | PHO  | 0       | 0.097647     |
| dunnli99wc  | TNO  | 0       | 0.095775     |

  
-> True COY meadoma99w rank: #1 (prob=0.3161)
```

Best model on Test: XGBoost_no_scaler

```
--- TEST Results (Year 10 Forecast) ---  
Model's Top 5 Predictions (XGBoost_no_scaler):  


| coachID    | tmID | won_coy | prob_winning |
|------------|------|---------|--------------|
| meadoma99w | ATL  | 1       | 0.566019     |
| whisejo99w | SAC  | 0       | 0.025072     |
| laimbbl01w | DET  | 0       | 0.022128     |
| donovan99w | NYL  | 0       | 0.021017     |
| bouceje01w | SAC  | 0       | 0.016921     |


```

Year 10 Actual COY (meadoma99w) Rank: #1
Predicted Probability: 0.5660

```
--- Generalization gap (model chosen on validation) ---  
AUC gap (test - validation): -0.0833  
AP gap (test - validation): -0.5000
```

```
--- Grid Search on TRAIN only (years < 9) for 5 models ---  
Positives in train: 6  
  
Training RandomForest no_scaler...  
Fitting 3 folds for each of 54 candidates, totalling 162 fits  
> Best ROC AUC (CV on train): 0.1867  
> Best Params: {'model_class_weight': 'balanced_subsample', 'model_max_depth': 5, 'model_min_samples_leaf': 4, 'model_n_estimators': 100}  
  
Training RandomForest scaled...  
Fitting 3 folds for each of 54 candidates, totalling 162 fits  
> Best ROC AUC (CV on train): 0.1867  
> Best Params: {'model_class_weight': 'balanced_subsample', 'model_max_depth': 5, 'model_min_samples_leaf': 4, 'model_n_estimators': 100}  
  
Training XGBoost no_scaler...  
Fitting 3 folds for each of 32 candidates, totalling 96 fits  
> Best ROC AUC (CV on train): 0.1767  
> Best Params: {'model_class_weight': 'balanced_subsample', 'model_max_depth': 3, 'model_min_samples_leaf': 4, 'model_n_estimators': 100, 'model_subsample': 1.0}  
  
Training XGBoost scaled...  
Fitting 3 folds for each of 32 candidates, totalling 96 fits  
> Best ROC AUC (CV on train): 0.1767  
> Best Params: {'model_class_weight': 'balanced_subsample', 'model_max_depth': 3, 'model_min_samples_leaf': 4, 'model_n_estimators': 100, 'model_subsample': 1.0}  
  
Training LogisticRegression scaled...  
Fitting 3 folds for each of 4 candidates, totalling 12 fits  
> Best ROC AUC (CV on train): 0.2800  
> Best Params: {'model_C': 10.0}
```

Despite very weak fixed-split cross-validation on the training years (all models < **0.5 ROC-AUC**), the tuned models still ranked the true COY as #1 in both the validation season (Year 9) and the test season (Year 10). With the rolling-window setup, performance becomes much stronger and more consistent, reaching perfect Year-10 results for the best models (**Test ROC-AUC = 1.00, Test AP = 1.00**).

```
--- Evaluating tuned models retrained on TRAIN+VAL ---  
  
--- Final comparison on Test (Year 10) ---  


|                           | Model    | Test AUC | Test AP | Top-5 hit (test) |
|---------------------------|----------|----------|---------|------------------|
| XGBoost_no_scaler         | 1.000000 | 1.000000 | 1       |                  |
| XGBoost_scaled            | 1.000000 | 1.000000 | 1       |                  |
| RandomForest_no_scaler    | 0.916667 | 0.500000 | 1       |                  |
| RandomForest_scaled       | 0.916667 | 0.500000 | 1       |                  |
| LogisticRegression_scaled | 0.500000 | 0.142857 | 0       |                  |


```

```
=== Soft-voting ensemble on Test (Year 10) - COY ===  
Ensemble Test AUC: 1.0000  
Ensemble Test AP: 1.0000  
  
--- Ensemble Top-5 candidates (Year 10) ---  


| coachID    | tmID | won_coy | prob_winning |
|------------|------|---------|--------------|
| meadoma99w | ATL  | 1       | 0.463010     |
| donovan99w | NYL  | 0       | 0.101080     |
| thibami99w | CON  | 0       | 0.079101     |
| bouceje01w | SAC  | 0       | 0.076698     |
| whisejo99w | SAC  | 0       | 0.052851     |

  
Ensemble Top-5 hit (test): 1  
  
Ensemble: Actual COY meadoma99w rank (Year 10): #1  
Winner prob (ensemble): 0.4630 (Top prob: 0.4630)
```

```
--- Validation performance (Year 9) for tuned models ---  
  
--- Model ranking on Validation (Year 9) ---  

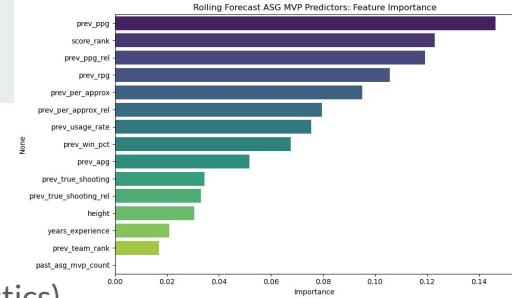

|                           | Model | Val AUC  | Val AP | Top-5 hit (val) |
|---------------------------|-------|----------|--------|-----------------|
| RandomForest_no_scaler    | 1.0   | 1.000000 | 1      |                 |
| RandomForest_scaled       | 1.0   | 1.000000 | 1      |                 |
| XGBoost_no_scaler         | 0.6   | 0.200000 | 1      |                 |
| XGBoost_scaled            | 0.6   | 0.200000 | 1      |                 |
| LogisticRegression_scaled | 0.4   | 0.142857 | 0      |                 |

  
Best model on validation: RandomForest_no_scaler  
  
--- Validation Results (Year 9) ---  
Model's Top 5 Predictions (best validation model):  


| coachID    | tmID | won_coy | prob_winning |
|------------|------|---------|--------------|
| thibami99w | CON  | 1       | 0.311327     |
| hugheda99w | SAS  | 0       | 0.237279     |
| aglerbr99w | SEA  | 0       | 0.160033     |
| dunnli99wc | IND  | 0       | 0.157948     |
| thompka99w | HOU  | 0       | 0.153296     |

  
Year 9 Actual COY (thibami99w) Rank: #1
```

All Star Game MVP Award



Goal: Predict the winner of a single exhibition game using only historical data (previous season statistics)

Feature Engineering Strategy: Lagged Performance, Legacy (previous won MVPs), Historical Bias, Score rank (own implemented feature, MVPs are often dominating scorers on top-seeded teams - based on previous team rank)

Modeling Approach: Rolling Forecast, Robustness (by excluding year 2004 and 2008 for Olympic years where no All Star Game was played)

Challenge: Model uses long-term data to predict short-term (game MVP)

Model Performance:
Overall ROC AUC: 0.636
after Rolling Model Validation

Final Prediction: Ranked the elite scorers of highest rank team last season very high

--- FINAL PREDICTION PHASE (Train on Years 2-9, Predict Year 10) ---

>>> Top 10 Pre-Season ASG MVP Candidates (Year 10):

	playerID	mvp_prob	prev_ppg	prev_ppg_rel	rank	past_asg_mvp_count		prev_usage_rate	score_rank
254	catchta01w	0.220933	13.280000	5.057291	1	0	254	0.532565	66.400000
592	hoffmeb01w	0.178101	10.363636	2.140928	1	0	592	0.394906	51.818182
1411	tauradi01w	0.140218	24.117647	15.894938	1	0	1411	0.711616	72.352941
1420	taylope01w	0.139450	17.794118	9.571409	1	0	1420	0.555564	53.382353
366	douglka01w	0.132747	15.636364	7.413655	1	1	366	0.549877	78.181818
803	leslili01w	0.117210	15.060606	6.837897	3	2	803	0.569292	90.363636
1467	thompti01w	0.117152	18.066667	9.843958	3	1	1467	0.581304	108.400000
1134	pondeca01w	0.110281	21.218750	12.996041	1	0	1134	0.746946	63.656250
819	littlca01w	0.096798	9.736842	1.514134	2	0	819	0.458824	68.157895
648	jacksla01w	0.092531	20.238095	12.015387	2	0	648	0.600346	141.666667

Final Test Set Predictions: Ranking

This slide shows the predicted rankings for Year 11. The predictions were generated using the model that achieved the best performance on the Year 10 test set under the fixed train-validation-test split, rather than the rolling-window approach, since the rolling runs were significantly more time-consuming and delivered only marginal differences in Year 10 test performance.

One specific issue in Year 11 is **TUL**, a team that did not exist in Year 10. Because there is no prior-season data to compute the required lag features, TUL was assigned an intentionally high predicted value so it would be ranked last within its conference.

Best Model in Testing (Year 10): Gradient Boosting

--- Predicted Rankings Year 11 ---

confID	tmID	year	predictedrankraw	predictedrankmodel
EA	IND	11	2.938715	1
EA	CON	11	3.242986	2
EA	ATL	11	4.080875	3
EA	NYL	11	4.905821	4
EA	WAS	11	4.906759	5
EA	CHI	11	4.921348	6
WE	SEA	11	3.015836	1
WE	LAS	11	3.167433	2
WE	SAS	11	3.585153	3
WE	PHO	11	3.895597	4
WE	MIN	11	5.101360	5
WE	TUL	11	1005.101360	6

Forced teams (missing features -> pushed to last):

tmID	confID	predictedrankraw	predictedrankmodel
TUL	WE	1005.10136	6



Final Test Set Predictions: Coach Change

Predicted coach change for:

- gilloje01w
- keyst99wc
- whisejo99w
- hugheda99w
- meadoma99w
- donovan99w
- thibami99w
- plankju99w

--- Final Test (Year 11) ---

>>> Coaches Most Likely to be Fired (Year 11):

	tmID	coachID	prob_fire	predicted_change
75	MIN	gilloje01w	0.583073	1
12	CHI	keyst99wc	0.571006	1
111	SAC	whisejo99w	0.556088	1
118	SAS	hugheda99w	0.537282	1
1	ATL	meadoma99w	0.531507	1
85	NYL	donovan99w	0.531237	1
23	CON	thibami99w	0.522841	1
141	WAS	plankju99w	0.520682	1
33	DET	mahorri01w	0.468051	0
128	SEA	aglerbr99w	0.462456	0
98	PHO	gaineco01w	0.456504	0
52	IND	dunnli99wc	0.455177	0
62	LAS	coopemi01w	0.445687	0

--- Final Classification Metrics (Year 11) ---

Test F1 Score (Class 1): 0.5714
Test Precision (Class 1): 0.5000
Test Recall (Class 1): 0.6667
Test Year 11 AUC: 0.5714

Final Test Set Predictions: Awards

	playerID	tmID	year	prob_winning	has_history
0	beviltu01w	SAS	11	0.253425	True
1	swoopsh01w	TUL	11	0.128375	True
2	mcwilta01w	MIN	11	0.084974	True
3	quinnno01w	LAS	11	0.075147	True
4	augusse01w	MIN	11	0.068235	True
5	birdsu01w	SEA	11	0.067839	True
6	youngso01w	SAS	11	0.064501	True
7	hodgero01w	SAS	11	0.051846	True
8	lehnish01w	ATL	11	0.048839	True
9	lawsoka01w	CON	11	0.045080	True

Kim Perrot
Sportsmanship
Award

	playerID	tmID	year	has_history	prob_winning
0	catchta01w	IND	11	True	0.552161
1	jacksla01w	SEA	11	True	0.374053
2	parkeca01w	LAS	11	True	0.257222
3	anosini01w	WAS	11	True	0.204694
4	douglka01w	IND	11	True	0.149444
5	dupreca01w	PHO	11	True	0.126824
6	tauradi01w	PHO	11	True	0.102719
7	perkiji01w	SAS	11	True	0.092217
8	smithta01w	IND	11	True	0.073079
9	lyttlsa01w	ATL	11	True	0.066018

Defensive Player
of the Year
Award

	playerID	tmID	year	has_history	prob_winning
0	catchta01w	IND	11	True	0.862577
1	hammobe01w	SAS	11	True	0.840078
2	cashsw01w	SEA	11	True	0.833035
3	suttota01w	IND	11	True	0.829869
4	youngso01w	SAS	11	True	0.751153
5	jonesas01w	CON	11	True	0.704462
6	douglka01w	IND	11	True	0.695954
7	parkeca01w	LAS	11	True	0.665016
8	rileyru01w	SAS	11	True	0.644152
9	thompti01w	LAS	11	True	0.612434
10	birdsu01w	SEA	11	True	0.608824
11	smithka01w	SEA	11	True	0.583562
12	braxtko01w	NYL	11	True	0.581435
13	taylope01w	PHO	11	True	0.571593
14	tauradi01w	PHO	11	True	0.566258
15	jacksla01w	SEA	11	True	0.551939
16	miltode01w	LAS	11	True	0.543033
17	zellosh01w	IND	11	True	0.541163
18	hoffmeb01w	LAS	11	True	0.533952
19	walkede01w	CON	11	True	0.528084

Finals MVP
Award

	coachID	tmID	year	has_history	prob_winning
0	whisejo99w	NYL	11	True	0.199958
1	hugheda99w	SAS	11	True	0.105113
2	gaineco01w	PHO	11	True	0.019009
3	laceytr99w	WAS	11	True	0.014116
4	dunnli99wc	IND	11	True	0.011966
5	thibami99w	CON	11	True	0.008865
6	gilloje01w	LAS	11	True	0.002946
7	aglerbr99w	SEA	11	True	0.001582
8	meadoma99w	ATL	11	True	0.000127
9	chatmda99w	CHI	11	False	NaN

Coach of the Year
Award



Final Test Set Predictions: All Star Game MVP Award

- Predicted catchta01w as MVP at the All Star Game

--- SECTION 5: FUTURE PREDICTION (Year 11) ---

Prediction: Top 10 All-Star Game MVP Candidates for Year 11:

	playerID	mvp_prob	prev_ppg	prev_team_rank	past_asg_mvp_count	score_rank
18	catchta01w	0.174485	15.058824	1	0	105.411765
27	dupreca01w	0.095354	15.705882	5	0	47.117647
26	douglka01w	0.093126	17.645161	1	1	123.516129
91	parkeca01w	0.081294	13.120000	3	0	65.600000
114	thompti01w	0.070088	12.970588	3	1	64.852941
68	lyttlsa01w	0.068451	13.029412	2	0	78.176471
44	holdscho1w	0.067252	13.920000	2	0	83.520000
129	youngso01w	0.066107	18.181818	4	0	72.727273
16	cashsw01w	0.063338	12.187500	2	1	73.125000
43	hoffmeh01w	0.054015	9.941176	1	0	69.588235

Final Test Set Predictions: Awards

Most Valuable Player Award

```
>>> TOP 10 PREDICTED CANDIDATES FOR YEAR 11:
```

	playerID	rank	prob	potential_ppg
106	pringla01w	1.0	0.055778	8.377358
51	holliqu01w	2.0	0.027939	13.603687
91	parisco01w	3.0	0.023983	13.031674
133	tolivkr01w	4.0	0.023974	19.119171
135	vaughki01w	5.0	0.017939	14.818182
53	hornbal01w	6.0	0.015972	9.021176
108	quinnno01w	7.0	0.015931	11.005382
40	gardike01w	8.0	0.011970	10.872483
59	januabr01w	9.0	0.010000	11.964861
114	sanfona01w	10.0	0.010000	11.565611

Most Improved Player Award

```
>>> Top 10 Pre-Season MVP Candidates (Year 11):
```

	playerID	mvp_prob	prev_ppg	prev_ppg_rel	team_rank_feature
1748	tauradi01w	0.707982	20.354839	11.506798	4.0
1761	youngso01w	0.517559	18.181818	9.333777	3.0
1700	jacksla01w	0.322561	19.192308	10.344267	1.0
1670	augusse01w	0.232139	21.000000	12.151959	5.0
1681	catchta01w	0.162466	15.058824	6.210783	1.0
1730	pondeca01w	0.116176	19.058824	10.210783	4.0
1731	powelni01w	0.113734	16.676471	7.828430	4.0
1704	jonesas01w	0.109861	16.695652	7.847611	2.0
1686	douglka01w	0.080019	17.645161	8.797120	1.0
1692	hammobe01w	0.072239	19.483871	10.635830	3.0

	playerID	prob	is_rookie	prev_ppg
12	bonnede01w	0.168	0	11.205882
127	thorner01w	0.046	0	6.205882
123	taylope01w	0.018	0	10.928571
65	lattaiv01w	0.010	0	6.083333
48	holliqu01w	0.006	0	4.823529
88	parisco01w	0.004	0	4.848485
104	quiglal01w	0.004	0	1.666667
34	elysh01w	0.002	0	6.823529
1	adamsda01w	0.000	1	4.773460
2	ajavoma01w	0.000	0	8.000000

Sixth Woman Award

```
Top 10 Year 11 ROY Candidates:
```

	playerID	tmID	prob	prev_team_win_pct	vacated_min_pct
	adairje01w	MIN	0.088810	0.411765	0.710940
	harriam01w	MIN	0.088810	0.411765	0.710940
	moorema01w	MIN	0.088810	0.411765	0.710940
	wrighmo01w	MIN	0.088810	0.411765	0.710940
	chestfe01w	NYL	0.083975	0.382353	0.705737
	brELAJE01w	NYL	0.083975	0.382353	0.705737
	montgal01w	NYL	0.083975	0.382353	0.705737
	colsosy01w	NYL	0.083975	0.382353	0.705737
	phillta02w	NYL	0.083975	0.382353	0.705737
	robinda01w	SAS	0.036100	0.441176	0.540222

Rookie of the Year Award



Conclusions, Limitations, Future Work

Overall, the project's goal was achieved: we obtained reasonable predictive performance for the ranking, coach changes, and awards, under the realistic constraint of not using any information from the season being predicted. Even though coach changes are inherently difficult to predict and the model only performs moderately better than random guessing, it still captures meaningful patterns that would be hard to spot manually.

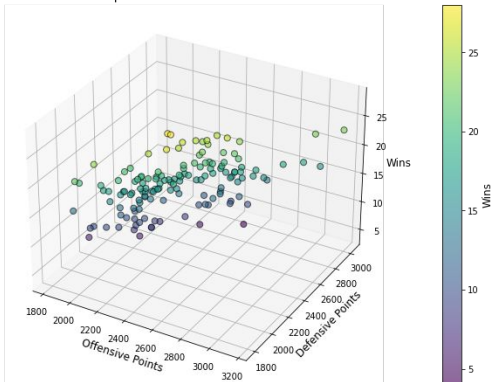
A key limitation is the relatively **small amount** and **granularity** of information available in the provided datasets, combined with the fact that the league itself was **unstable** during this period, with teams and coaches frequently changing, folding or relocating. This contrasts with real-world predictive systems, which typically rely on richer and more stable data to support stronger conclusions.

For the **All-Star Game MVP**, even though a prediction model was implemented, this award remains difficult to forecast: it depends first on which players are selected to the game through fan, player and media voting, and then on who happens to perform best in a single exhibition game, for which no game-level data was available in our dataset. As a result, the model can only approximate voting and game context indirectly via season-long performance, which is not always aligned with how All-Star MVPs are chosen in practice.

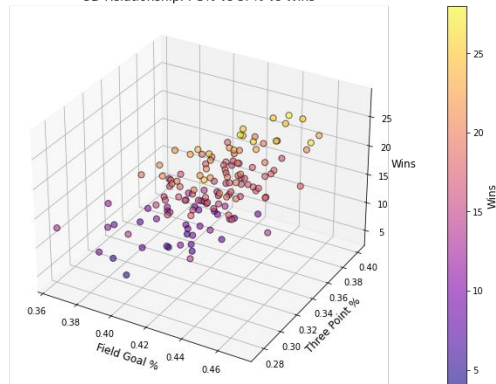
As future work, the predictive performance could be improved by **extending the dataset** with more seasons and richer covariates. In the specific case of the coach change problem, including information about contract duration and terms would likely be a key driver, as it strongly constraints when a coach is realistically able or likely to leave a team.

Appendix A: Teams Analysis

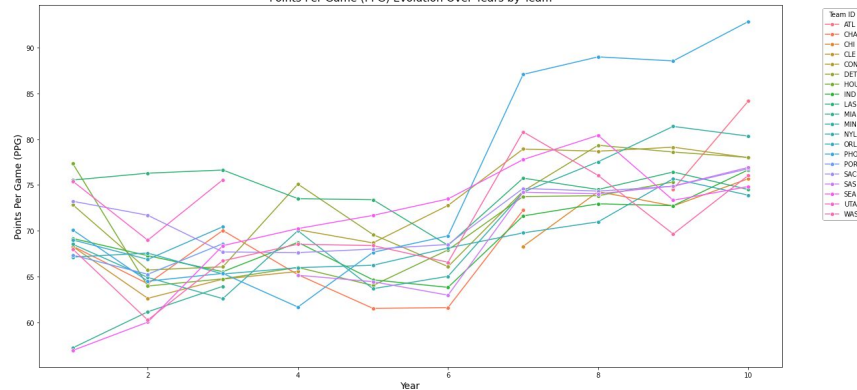
3D Relationship: Offensive vs Defensive Points vs Wins



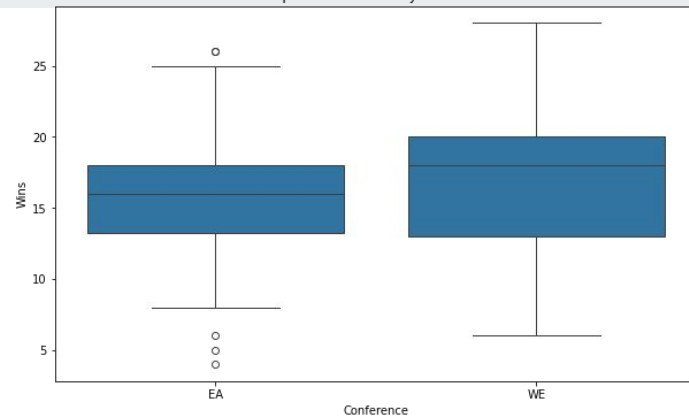
3D Relationship: FG% vs 3P% vs Wins



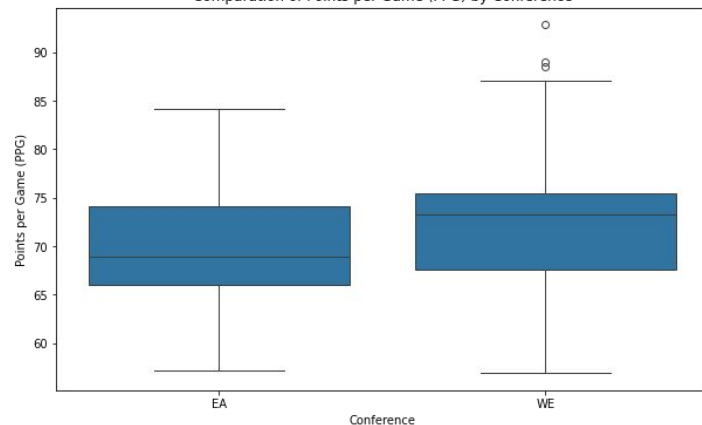
Points Per Game (PPG) Evolution Over Years by Team



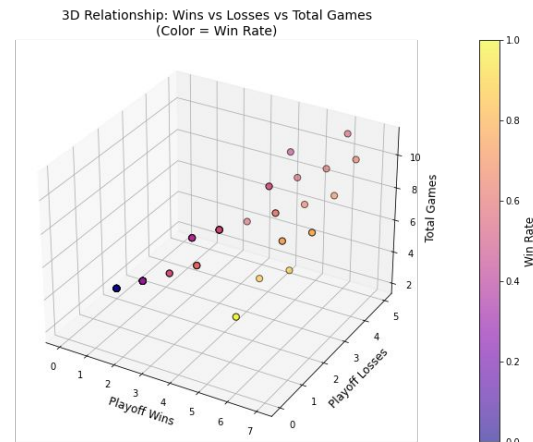
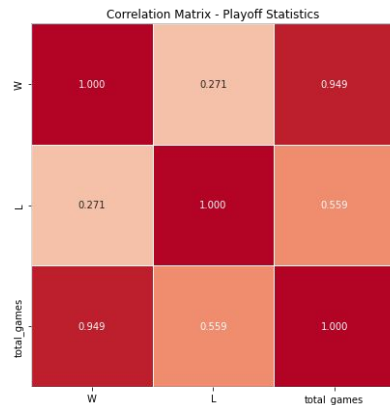
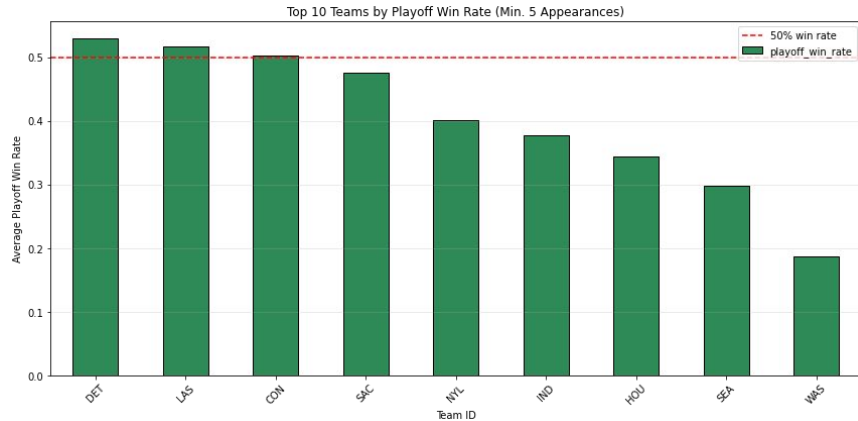
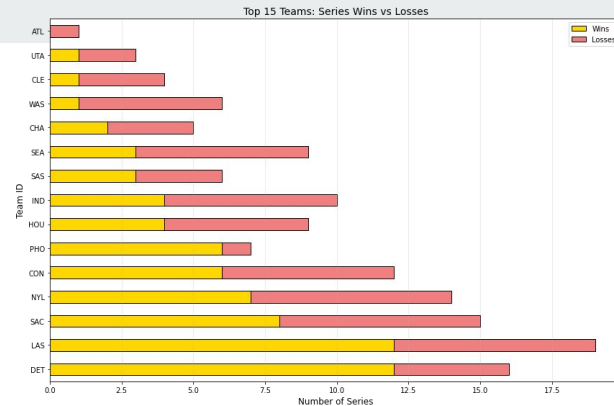
Comparison of Wins by Conference



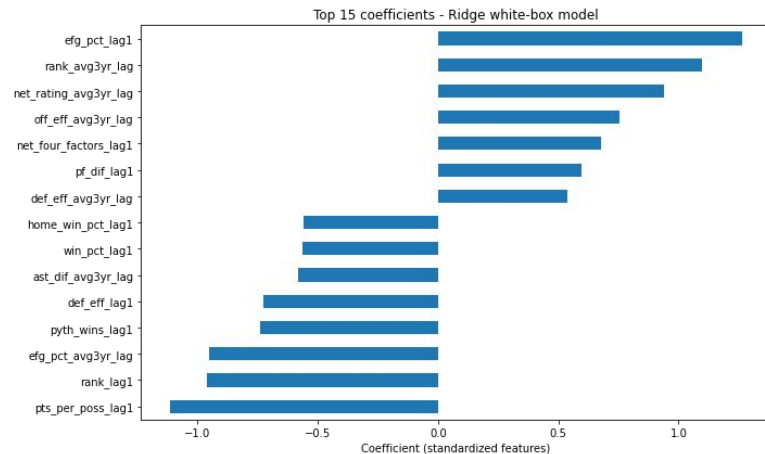
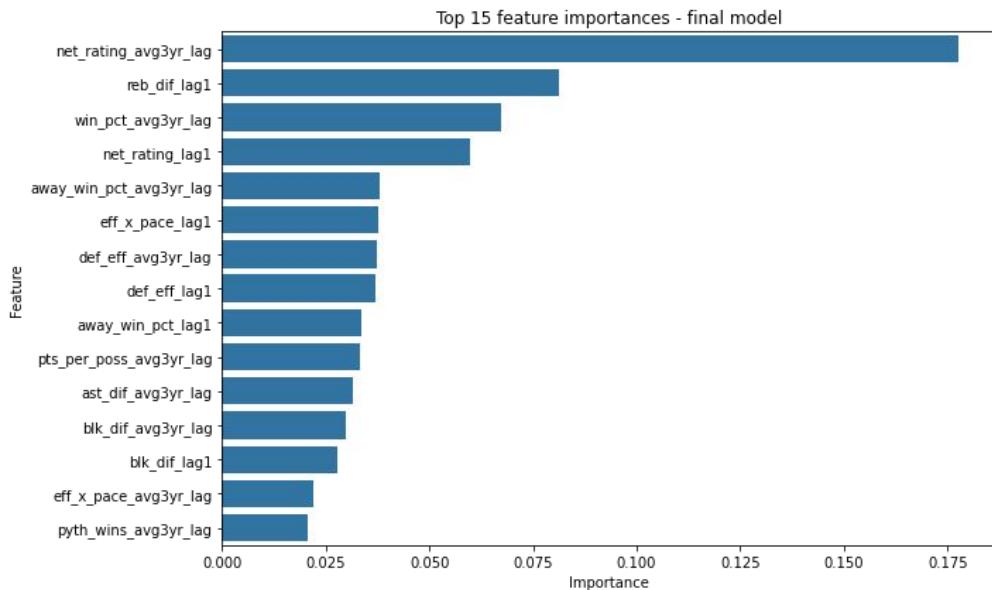
Comparison of Points per Game (PPG) by Conference



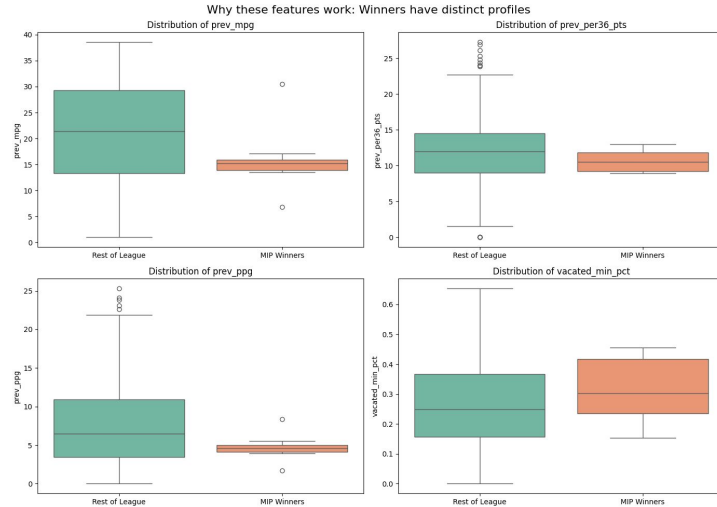
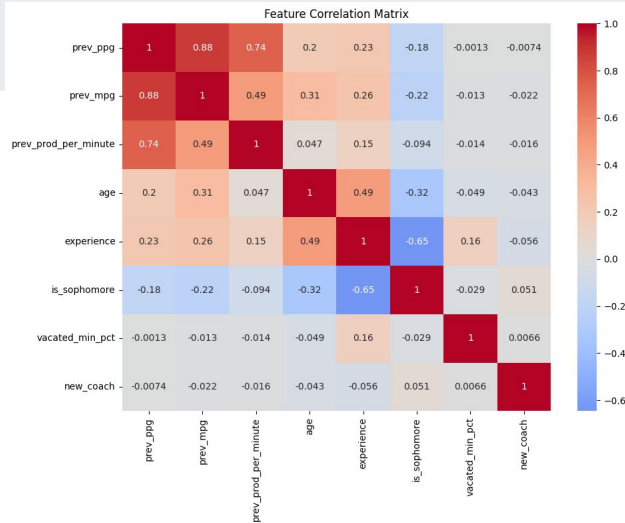
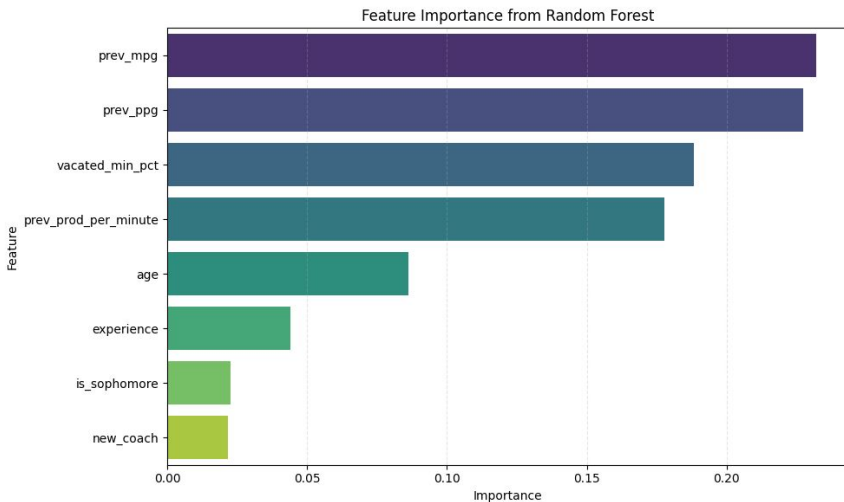
Appendix B: Playoffs Analysis



Appendix C: Feature Importance and White-box Model for the Ranking Problem



Appendix D: Data Analysis, Most Improved Player Award



Appendix E: Data Analysis, Most Valuable Player Award

